high-intensity values.

- It improves the visibility of objects in poorly illuminated areas.

Effect:- The dark parking lot becomes brighter, and objects become easier to distinguish.

3) Fingerprint Recognition

Transformation: contrast stretching

Reason:- It increases the difference between the gray levels of fingerprint ridges and valleys.

* It improves the visibility of fine ridge patterns.
* It makes fingerprint details easier to analyze
* It helps separate foreground ridge patterns from the background.
* Logarithmic transformation may not provide the desired ridge-valley contrast in every fingerprint image.

Effect:- contrast stretching produces a clearer finger-print image with more distinguishable ridges and valleys.

Activity - 1

1) Medical Image Enhancement

A doctor receives an x-ray image in which the bones are visible, but the darker tissues have very poor visibility.

a) I recommand power-law (Gamma) Transformation with the value less than 1.

Reason:- It increases the brightness of darker regions in the x-ray images

* It improves the visibility of soft tissues and other hidden details
* It helps distinguish structure with different intensity levels.
* The enhanced image makes regions easier for doctors to examine.

Effect:- The x-ray becomes brighter, and details in darker tissues become more visible.

2) Night-Time Surveillance

Transformation:- I recommand Logarithmic Transformation

Reason:- It is useful when most pixel values are concentrated in the lower intensity range

* It expands low-intensity values and compresses

high-intensity valu
- It improves the visi
illuminated areas.

Effect:- The dark
objects become ea

3) fingerprint Reco

Transformation:
Reason:- It inc
gray levels of
* It improves
* It makes fin
* It helps
from the ba
* Logarithmic
the desired
fingerprint
Effect:- con
print image
valleys.